Name of Institute: Indus Institute of Technology & Engineering
Name of Faculty: Prof. Abhishek Vaghela

Course code: CE0404
Course name: Computer Organization and Architecture

Pre-requisites: Digital Electronics
Credit points: 3
Offered Semester: IV

Course Coordinator
Full name: Prof. Abhishek Vaghela
Department with sitting location: B225, Second floor Bhanwar Building.
Telephone: 3204
Email: abhishekvaghela.ce@indusuni.ac.in

Course lecturers
Full Name: Prof. Abhishek Vaghela
Department with sitting location: B225, Second floor Bhanwar Building.

Students will be contacted throughout the session via mail and Google classroom with important information relating to this course.

Course Objectives:

1. To conceptualize the basics of organizational and architectural issues of a digital computer.
2. To analyze performance issues in processor and memory design of a digital computer.
3. To understand various data transfer techniques in digital computer.
4. To analyze processor performance improvement using parallelism.

Course Outcomes (CO)

After successful completion of the course, student will able to:

1. To understand and describe the basics of various architectural units of the Computer System.
2. To apply the knowledge of combined and sequential logical circuits to mimic a simple computer architecture.
3. To demonstrate the simulations for basic computer operations.
4. To recognize the importance of parallelism in computer architecture.
5. To understand the basic working of various memory system.
6. This course is the bridge between digital electronics and microprocessor.

Course Outline

UNIT-I **[12 hours]**

Basic Computer Organization and Data Representation: Overview of computer organization, CPU organization components classification of computer architecture, Register, types of register, Bus, types of bus, quantitative techniques in computer design,measuring and reporting performance, Amdahl's Law, number system, fixed point representation, floating point representation.

UNIT-II **[12 hours]**

Instruction Set Architecture: Instruction, instruction format, instruction execution cycle, instruction timing cycles, ISA, ISA design issue, types of ISA, Accumulator organization, stack organization, memory-memory organization, register memory organization, register-register organization, addressing mode, RISC, CISC,8085 microprocessors, Architecture, Operations of 8085, Instruction Set for 8085, Addressing Modes of 8085.

UNIT-III **[12 hours]**

Pipeline : What is pipeline, Flynn's taxonomy for Parallel Processing, types of pipeline, Arithmetic Pipeline, Instruction Pipeline, measuring performance of pipeline, types of hazard, structural hazard, data hazard, control hazard, method for avoiding hazard, Vector Processing, Array Processors, Parallel Processing

UNIT-IV **[12 hours]**

Memory Organization: Memory Hierarchy, types of memory, Main Memory, Auxiliary Memory, Inclusion, Coherence and locality properties, Cache Memory, Organization of cache memory, performance of cache memory with examples, cache memory mapping algorithms, cache optimization techniques, Virtual Memory, demand paging, page replacement algorithms.

Method of delivery

Chalk and Board, PowerPoint Presentation

Study time

3 hrs theory

CO-PO Mapping (PO: Program Outcomes)

- **Engineering Graduates will be able to:**
- **PO1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO3. Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4. Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5. Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **PO**6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **PO7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8. Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9. Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **PO11. Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's ownwork, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12. Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	1	3	1	1	-	-	1	2	3	1	1
CO2	-	1	3	1	1	-	-	1	2	3	1	1
CO3	-	1	3	1	1	-	-	1	2	3	1	1
CO4	-	1	3	1	1	-	-	1	2	3	1	1
CO5	-	1	3	1	1	-	-	1	2	3	1	1
CO6	-	1	3	1	1	-	-	1	2	3	1	1

Bloom's Taxonomy and Knowledge retention (For reference)

(Bloom's taxonomy has been given for reference)

Figure 1: Bloom's Taxonomy

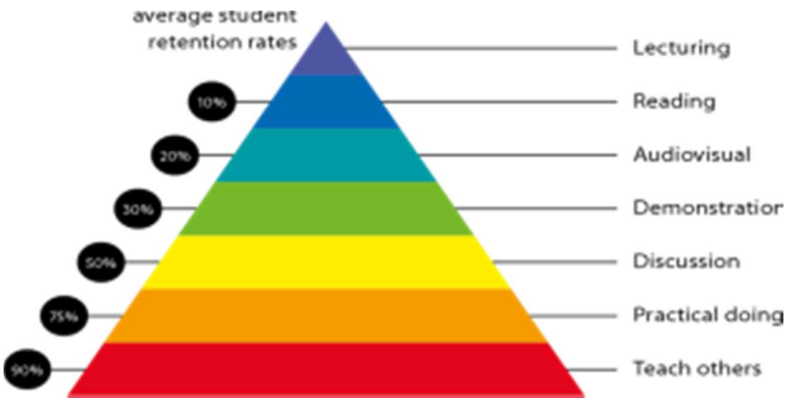


Figure 2: Knowledge retention

Graduate Qualities and Capabilities covered

(Qualities graduates harness crediting this Course)

General Graduate Qualities	Specific Department of ___Graduate Capabilities
Informed Have a sound knowledge of an area of study or profession and understand its current issues, locally and internationally. Know how to apply this knowledge. Understand how an area of study has developed and how it relates to other areas.	**1 Professional knowledge, grounding & awareness**
Independent learners Engage with new ideas and ways of thinking and critically analyze issues. Seek to extend knowledge through ongoing research, enquiry and reflection. Find and evaluate information, using a variety of sources and technologies. Acknowledge the work and ideas of others.	**2 Information literacy, gathering & processing**
Problem solvers Take on challenges and opportunities. Apply creative, logical and critical thinking skills to respond effectively. Make and implement decisions. Be flexible, thorough, innovative and aim for high standards.	**4 Problem solving skills**
Effective communicators Articulate ideas and convey them	**5 Written communication** **6 Oral communication**

effectively using a range of media. Work collaboratively and engage with people in different settings. Recognize how culture can shape communication.	**7 Teamwork**
Responsible Understand how decisions can affect others and make ethically informed choices. Appreciate and respect diversity. Act with integrity as part of local, national, global and professional communities.	**10 Sustainability, societal & environmental impact**

Lecture times

(Give lecture times in the format below)

Lecture Monday	**10:50 AM to 11:45 AM(4th IT-C)**
Lecture Wednesday	**09:55 AM to 10:50 AM(4th IT-C)**
	01:20 PM to 02:15 PM(4th CE-B)
Lecture Thursday	**09:55 AM to 10:50 AM(4th CE-B)**
	11:45 AM to 12:40 PM(4th IT-C)
Lecture Friday	**09:55 AM to 10:50 AM(4th CE-B)**

The University norms state that it is the responsibility of students to attend all lectures, tutorials, seminars and practical work as stipulated in the course outline. Minimum attendance requirement as per university norms is compulsory for being eligible for semester examinations.

Text Books:

1. Computer system Architecture Mano, M. Morris Pearson publication
2. Computer Architecture and Organization Ghoshal, Subrata Pearson publication

Reference Books:

1. Carl Hamacher - Computer Organization, Fifth Edition, Tata McGraw Hill.
2. John P Hayes - Computer Architecture & Organization–McGraw Hill
3. William Stallings - Computer Organization and Architecture, Seventh Edition, Pearson Education

Web Resources:

1. https://nptel.ac.in/courses/106/105/106105163
2. https://www.youtube.com/channel/UC2GUBG_WsP0OO5tXXocwp3Q/videos

ASSESSMENT GUIDELINES

Your final course mark will be calculated from the following:

CIE-Theory (60 Marks) **Assignment:10** **Attendance + Quiz: 10** **Mid Sem Exam:40**
ESE-Theory- 40
Total: 100 Marks

SUPPLEMENTARY ASSESSMENT

Students who receive an overall mark less than 40% in internal component or less than 40% in the end semester will be considered for supplementary assessment in the respective components (i.e internal component or end semester) of semester concerned. Students must make themselves available during the supplementary examination period to take up the respective components (internal component or end semester) and need to obtain the required minimum 40% marks to clear the concerned components.

Practical Work Report/Laboratory Report:

A report on the practical work is due the subsequent week after completion of the class by each group.

Late Work

Late assignments will not be accepted without supporting documentation. Late submission of the reports will result in a deduction of -% of the maximum mark per calendar day

Format

All assignments must be presented in a neat, legible format with all information sources correctly referenced. **Assignment material handed in throughout the session that is not neat and legible will not be marked and will be returned to the student.**

Retention of Written Work

Written assessment work will be retained by the Course coordinator/lecturer for two weeks after marking to be collected by the students.

University and Faculty Policies

Students should make themselves aware of the University and/or Faculty Policies regarding plagiarism, special consideration, supplementary examinations and other educational issues and student matters.

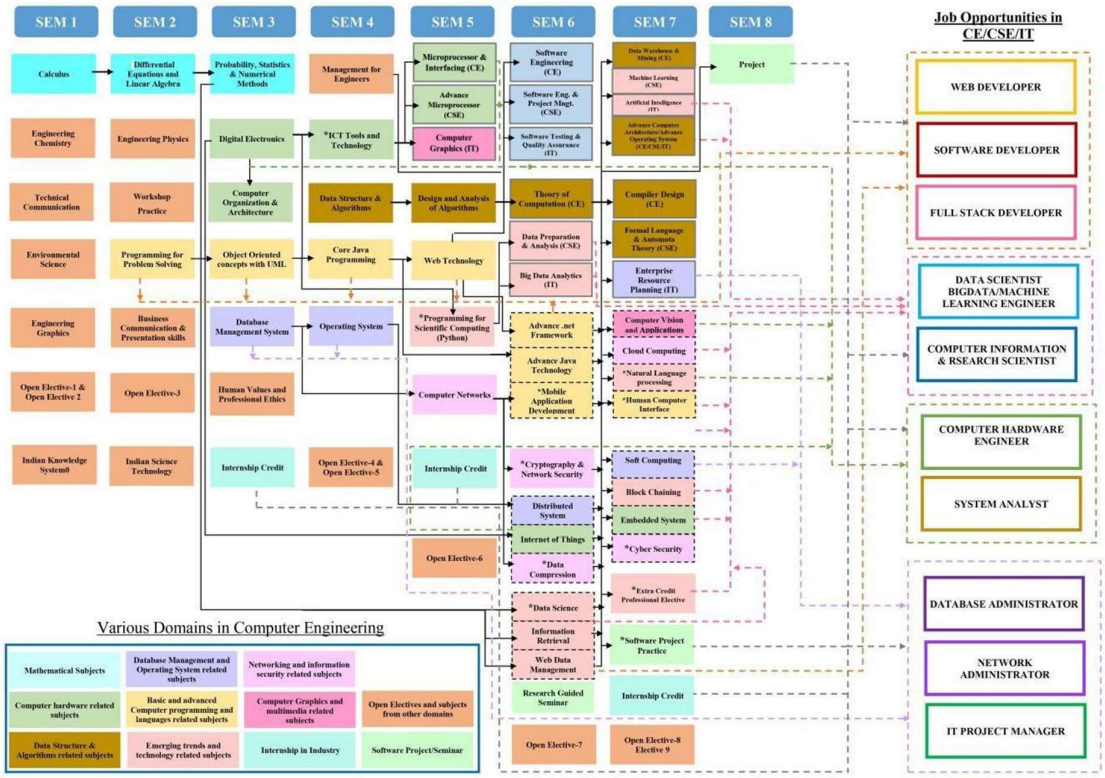
Plagiarism - Plagiarism is not acceptable and may result in the imposition of severe penalties. Plagiarism is the use of another person's work, or idea, as if it is his or her own - if you have any doubts at all on what constitutes plagiarism, please consult your Course coordinator or lecturer. Plagiarism will be penalized severely.

Do not copy the work of other students.
Do not share your work with other students (except where required for a group activity or assessment.

Course schedule

	Week #	Topic & contents	CO Addressed	Teaching Learning Activity (TLA)
	Week 1	Overview of computer Organization, CPU organization components classification of Computer architecture	I	Discussion, Presentation, Chalk & Board
	Week 2	Register, types of register, Bus, types of bus	I	Presentation, Chalk & Board

	Week 3	quantitative techniques in computer design, measuring and reporting performance	I	Presentation, Chalk & Board
	Week 4	Amdahl's Law, number system, Fixed point representation, floating point representation	II	Presentation, Chalk & Board
	Week 5	Instruction, instruction format, instruction execution cycle, instruction timing cycles	II	Presentation, Chalk & Board
	Week 6	ISA, ISA design issue, types of ISA, Accumulator organization, stack organization	IV	Presentation, Chalk & Board
	Week 7	memory-memory organization, register memory organization, register-register organization, addressing mode	III	Presentation, Chalk & Board, Demonstration
	Week 8	RISC, CISC,8085 microprocessors, Architecture, Operations of 8085, Instruction Set for 8085, Addressing Modes of 8085.	III,IV	Presentation, Chalk & Board, Demonstration
	Week 9	What is pipeline, Flynn's taxonomy for Parallel Processing, types of pipeline	II	Presentation, Chalk & Board
	Week 10	Arithmetic Pipeline, Instruction Pipeline	IV	Presentation, Chalk & Board
	Week 11	measuring performance of pipeline, types of hazard, structural hazard, data hazard, control hazard, method for avoiding hazard	IV	Presentation, Chalk & Board
	Week 12	Vector Processing, Array Processors, Parallel Processing	V,VI	Presentation, Chalk & Board
	Week 13	Memory Hierarchy, types of memory, Main Memory, Auxiliary Memory	V,VI	Presentation, Chalk & Board
	Week 14	Inclusion, Coherence and locality properties, Cache Memory, Organization of cache memory	V,VI	Presentation, Chalk & Board

	Week 15	Performance of cache memory with examples, cache memory mapping algorithms	V,VI	Presentation, Chalk & Board
	Week 16	Cache optimization techniques, Virtual Memory, demand paging, page replacement algorithms	V,VI	Presentation, Chalk & Board

COMPUTER ENGINEERING DEPARTMENT COURSE DEPENDANCY CHART